Course Name: DATABASE MANAGEMENT SYSTEM

Course Outcome

CO1: Understand the concept of database management systems and Relational database.
CO2: Identify the various data model used in database design.
CO3: Design conceptual models of a database using ER modeling for real life applications and construct queries in Relational Algebra.
CO4: Create and populate a RDBMS for a real life application, with constraints and keys using SQL.
CO5: Select the information from a database by formulating complex queries in SQL.
CO6: Analyze the existing design of a database schema and apply concepts of normalization to design an optimal database.
CO7: Discuss indexing mechanisms for efficient retrieval of information from a database.
CO8: Discuss recovery system and be familiar with introduction to web database, distributed databases.

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Mid Term Examination, Odd Semester 2023-24
B. TECH. (CS), Year: II, Semester: III
Subject Code: BCSC 1003
Subject Name: Database Management System

Time: 2 Hours **Maximum Marks: 30**

Instruction for students:
1. Mention clearly if any assumptions are being made.
2. No 'B' sheet will be given.

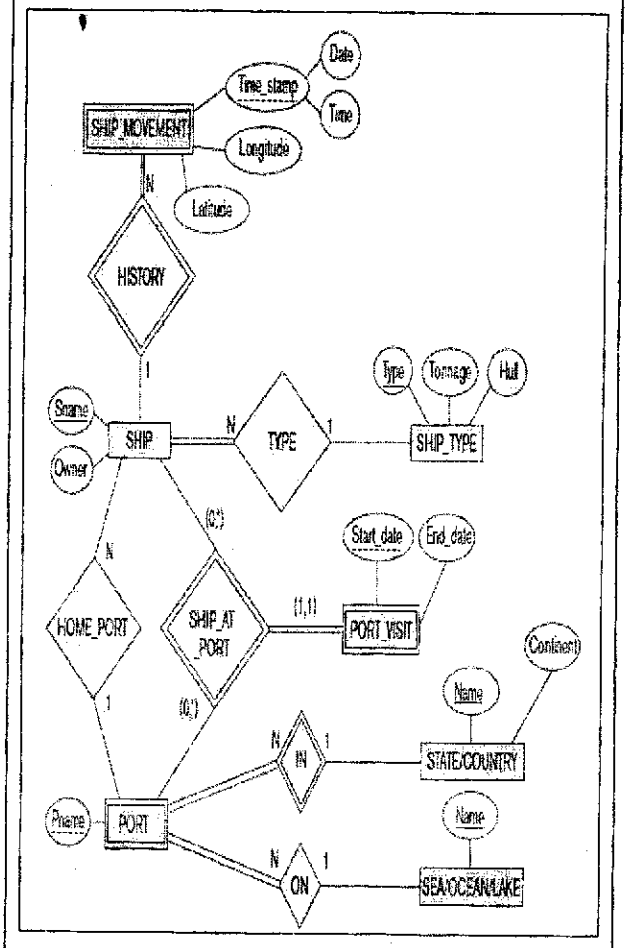
Section – A

Attempt All Questions 3 X 5 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
1	*"DBMS is preferred by industries over File Systems".* Justify your answer with proper reasons and scenario.	3	1	U	C
2	Assume that you are appointed in your dream company as a database designer and got a project to work on backend implementation. Your job profile includes designing a complete database for a web based project. Identify and briefly explain all the phases you need to handle during the database development life cycle during this project.	3	1	U	C
3	Write short note on: (a) Hierarchical Data Model, (b) Network Data Model, and (c) Relational Data Model.	3	2	R	F
4	Define RENAME (ρ) operator in details. Is it possible to apply SELECT (σ) operator and PROJECT (π) operator in a single SQL syntax? If yes, then explain how. If no, then explain why.	3	3 & 4	An	C
5	Consider a relation schema R = (A, B, C, D, E, F) on which the following functional dependencies hold: {A → B, BC →D, E → C, D → A} Identify the Candidate Key(s) of R.	3	4	E	P

Section – B

Attempt All Questions 5 X 3 = 15 Marks

No.	Detail of Question	Marks	CO	BL	KL
6	Consider the ER diagram given below & convert it into all possible tables: 	5	3	An	P
7	Suppose following tables are created in a database: College (cName, state, enrollment) Student (sID, sName, GPA, sizeHS) Apply (sID, cName, major)	5	5	C	P

	Now, write the SQL syntax for the following queries: a. Add cName as primary key in the College table. b. Make cName in Apply table as foreign key referring college table. c. Rename column state to address in the College table. d. Increase size of address attribute in College table from 15 to 25. e. Drop primary key from the College table.				
8	Consider that a relation STUDENT_PERFORMANCE (Name, CourseNo, RollNo, Grade) has the following functional dependencies: {Name, CourseNo → Grade, RollNo, CourseNo → Grade, Name → RollNo, RollNo → Name} Determine the highest normal form of this relation.	5	6	E	C